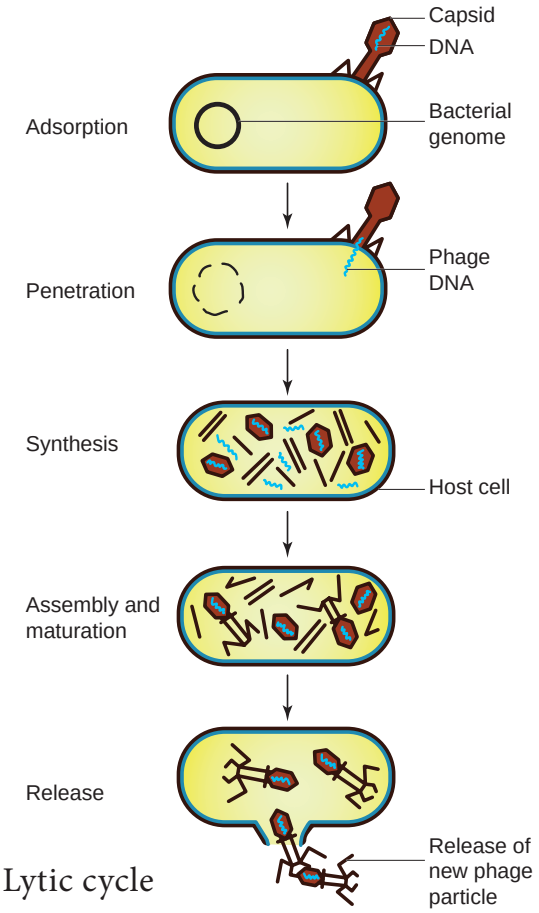
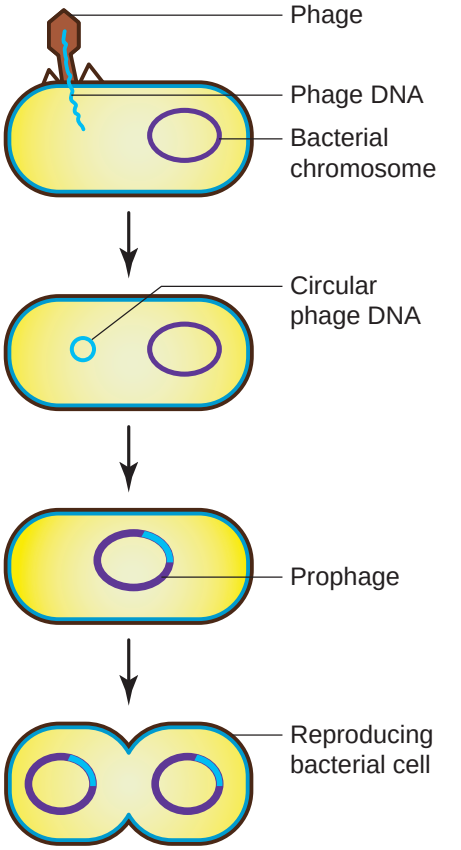
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(a) Lytic cycle




(b) Lysogenic cycle

**Figure 1.5:** Multiplication cycle of phage,

assembling the phage particles is known as **maturation**. After 20 minutes of infection about 300 new phages are assembled.

**(v) Release**

The phage particle gets accumulated inside the host cell and are released by the lysis of host cell wall.

### *b. Lysogenic Cycle*

In the lysogenic cycle the phage DNA gets integrated into host DNA and gets multiplied along with nucleic acid of the host. No independent viral particle is formed (Figure 1.5b).

As soon as the phage injects its linear DNA into the host cell, it becomes circular and integrates into the bacterial chromosome by recombination. The integrated phage DNA is now called **prophage.** The activity of the prophage gene is repressed by two repressor proteins which are synthesized by phage genes. This checks the synthesis of new phages within the host cell. However, each time the bacterial cell divides, the prophage multiplies along with the bacterial chromosome. On exposure to UV radiation and chemicals the excision of phage DNA may occur and results in lytic cycle.

**Virion** is an intact infective virus particle which is non-replicating outside a host cell.

**Viroid** is a circular molecule of ssRNA without a capsid and was discovered by T.O.Diener in the year 1971. The RNA of viroid has low molecular weight. Viroids cause citrus exocortis and potato spindle tuber disease in plants.

**Virusoids** were discovered by J.W.Randles and Co-workers in 1981. They are the small circular RNAs which